

S06 MINIMUM BACKGROUND SOUND | O (MAX : 2 PT)

Intent : Increase acoustical privacy within and between occupied spaces.

Summary : This WELL feature requires the use of dedicated artificial sound to uniformly increase speech privacy between occupied spaces.

Issue : Speech privacy can be described as the signal-to-noise ratio, or sound blocking and sound cover, of perceptible speech reaching a listener.¹ Ambient background sound, which can include artificial sound sources, can increase speech privacy to comfortable levels.^{2,3} However, ambient background noise provided by HVAC systems alone is shown to fluctuate by zone and by as much as 15 dBA or more in a given year.⁴ Some HVAC systems, like chilled beams, offer little to no ambient sound contribution and result in quiet interiors, where speech is readily intelligible and distracting. In offices and patient rooms where speech privacy is critical between regularly occupied spaces, a combination of reliable background sound is needed where partitions do not perform above a minimum STC-30.⁵

Solutions : Consistent background sound can be delivered to a space using an adjustable array of loudspeakers located such that sound is uniformly distributed.^{6,7} These systems are programmed to output a sound source similar to the sound of air flowing through HVAC. The sound source is broad-band and designed to elevate ambient background sound level to improve the signal-to-noise ratio and reduce speech perception between enclosed and open environments.^{8,9}

PART 1 PROVIDE MINIMUM BACKGROUND SOUND (MAX : 1 PT)

For Office Spaces:

The following requirements are met:

- a. A sound masking system is installed in open areas and enclosed rooms designated as quiet zones, circulation zones and in areas where workstations are present.
- b. The sound masking system produces a 1/3 octave band adjustable output signal and minimum frequency spectrum of 100 Hz to 5 kHz.
- c. The sound masking system is commissioned such that the following sound pressure levels are not exceeded:
 1. Open areas designated as quiet zones, circulation zones and areas where workstations are present: 48 dBA.
 2. Enclosed rooms labeled as quiet zones: 42 dBA.
- d. The sound masking system is verified by a professional sound masking system installer in accordance with ASTM 1573-18 or equivalent standard.

PART 2 PROVIDE ENHANCED SPEECH REDUCTION (MAX : 1 PT)

For All Spaces:

The project achieves the following features and parts:

- a. Achieve two points in either Feature S03 Part 2 - Achieve Sound Isolation at Walls or Feature S05 Part 1 – Implement Sound Reducing Surfaces.
- b. Feature S06 Part 1 – Provide Minimum Background Sound.

REFERENCES

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